

VR-Enhance: Aiding human speech and sensorimotor skills using Virtual Reality

Using AI to aid Speech, Occupational and Physical Therapy

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ABSTRACT

Stroke remains one of the major causes for most language and functional disabilities, but this disease is not the only cause for such deficits. Current rehabilitation programs struggle to keep up with increasing demands for therapy, each and every day. This paper proposes VR-Enhance, a novel VR-based rehabilitation game system which uses Deep Learning techniques for recognising both gestures and speech, in a single application. The solution aims to provide an alternate means of therapy by allowing patients to independently improve their speech and physical abilities, with minimal to no guidance from therapists.

Keywords

Rehabilitation, Virtual Reality, Speech Recognition, Gesture Recognition, Deep Learning, Multimodality

1. INTRODUCTION

1.1 Problem Definition

Today, billions of people require rehabilitation services, and the number keeps on growing [1]. The overarching goal of rehabilitation programs based on speech, occupational or physical therapy is to restore the person's ability to participate in normal life roles with as much independence as possible. It is therefore paramount that such services be made more accessible, all around the world.

The lengthy therapy process would involve guided limb and vocal manipulation, targeting specific regions of the patient's body. For this process to be effective, rehabilitation must be early, intensive and repetitive [2], which may result in challenges related to exhaustion and a lack of motivation for the patient. The induced psychological impact must not be underestimated, as many patients are also shown to regress into depression [3]. In addition, such therapeutic programs are usually very time consuming, leading to high societal costs and large dependency on the expertise of the therapist, whom may not have enough time to treat all assigned patients in a conventional manner. In such cases, the patient would most likely be required to perform therapy multiple times at home, with no supervision or real-time guidance. Such programs may lack emphasis on intense language paradigms in speech domains, and are rarely adapted to the personal needs of each patient [4]. It is therefore imperative to determine alternate means of rehabilitation which could fulfil and mitigate the above limitations, to both aid patients and the therapists.

1.2 Motivation

Virtual reality-based rehabilitation allows the integration of conventional rehabilitation with interactive virtual environments (VE) that can be structured according to the needs of the patient. There has been extensive research comparing Virtual Reality (VR) therapies to conventional methods as such technologies have become more readily available and affordable. The use of VR in therapeutic domains is proven to match the standard therapy approaches in some cases [2, 5] and even perform better in other cases [6, 7]. In one study which compares VR based training to conventional therapy (CT) [8], patients who were less impaired showed higher improvements in favour of the more severely impaired patients. Those that were severely impaired however, showed similar results to those that had undergone CT.

Findings have also changed the approach towards traditional language rehabilitation which reinforces the idea that better results could be achieved through stimulation of multiple brain regions [9, 10]. As suggested in [4], for speech therapy to be effective, there must be an interaction between the linguistic neural system and cortical sensorimotor circuits for providing an ideal basis for language comprehension and production. VR technologies are tools that could encapsulate the above requirements by providing a multimodal system for recognising speech and gesture simultaneously.

Currently, there lacks a VR solution which captures the requirements of the 3 different rehabilitation domains into one application. Research has been made on VR technologies using speech recognition [11, 12] and gesture recognition [13, 14], but there has yet to exist a solution which combines the two together for VR rehabilitation.

2. AIMS AND OBJECTIVES

The aim of this report is to investigate how AI can improve a VR based rehabilitation application set in a game environment, to target speech and gesture treatment. This will be accomplished by fulfilling the following objectives:

- **Objective (O1)** : Provide accurate identification of user gestures using a Deep Neural Network (DNN), to increase precision.
- **Objective (O2)** : Identify user speech using a separate DNN, which works simultaneously to that of the gestures.
- **Objective (O3)** : Provide a dynamic experience by determining the colour, type and size of spells along with

the intensity of haptic feedback by using fuzzy logic; of which is determined through confidence values returned by the DNN results.

- **Objective (O4)** : Ensure gradual user improvement by building an informative graphical interface situated in the user's game environment to show recognition result percentages and a history of scores in-game.

3. BACKGROUND RESEARCH

3.1 Artificial Neural Networks

An artificial neural network (ANN) is a machine learning paradigm designed to simulate the interconnecting neurons of the human brain through layers of nodes. ANNs are the foundation of AI which are capable of self-learning to represent complex functions which produce better results with larger data sets. These structures are built as an interconnected set of layers, where each layer is made up of a number of neurons. The most simple ANN consists of an input layer, a single hidden layer, and an output layer, all connected in the order in which they are listed. Each neuron within the network takes a real-valued input and produces a real-valued output to be carried over to the neuron in the following layer.

3.2 Deep Learning

Whereas standard neural networks usually consist of not more than 2 to 3 hidden layers, Deep Learning (DL) allows for more processing layers to be present within a deep neural network (DNN) [15]. The need for DNNs came about when larger inputs such as images had to be processed. ANNs are incapable of processing such large inputs due to their shallow nature and lack of connections among neurons in the same layer. Therefore, by increasing the number of hidden layers by a considerable amount, networks were now capable of modelling more complex, non-linear relationships. DL has drastically improved the state of the art in AI technologies, and continues to bring about breakthroughs in domains such as those of speech and gesture recognition.

3.3 Speech Recognition

Automatic Speech Recognition (ASR), also known as speech-to-text (STT) is the domain which focuses on processing human speech into written format. While it is commonly confused with voice recognition and natural language processing (NLP), STT focuses on receiving speech input and translating this input into text. Voice recognition on the other hand seeks to identify an individual from the user's voice; Whilst NLP is an AI domain which focuses on human computer interaction (HCI), using STT.

3.4 Gesture Recognition

Gesture Recognition focuses on the interpretation of human gestures which are a form of nonverbal communication used in everyday life. Gesture understanding is already utilised in many of today's applications and research is still being made to enhance domains like those of HCI [16].

It is important to note that, gestures can include the entire human body, however, this study will mainly focus on hand gestures. These greatly define an individual's body language and can be conveyed through varying finger positions and the shape of the hand. Hand gestures can either be classified as being static or dynamic [17, 18]. Static gestures like the

peace sign, refer to the hand and finger positions at a stationary state, whilst dynamic gestures refer to a sequence of hand movements which would represent the overall gesture, such as a military salute.

4. LITERATURE REVIEW

In the previous section, the main components applied within this study are discussed and explained briefly. This chapter focuses on providing a richer overview on the current state-of-the-art research and applications, that are relevant to the aims of this thesis. The application of VR in therapeutic domains is reviewed and their usefulness in such fields is determined. A review of speech and gesture recognition approaches is also carried out. Following these, the discussion ends on the combination of the two modules to tackle the idea of multi-modal interaction to determine its effectiveness.

4.1 VR Rehabilitation

By developing a VR rehabilitation tool [4], Grechuta et al. try to address the question of whether rehabilitation, more specifically that of aphasia, is effective when designed within a VE. Applying the intensive language action therapy paradigm to the solution, results show that not only did the participants improve in core language and communication skills, but these were also satisfied with the system. Despite their achievements, the authors still seek room for improvement and claim that such a system could be enhanced by handling hand-tracking for allowing simultaneous use of speech and motor actions. Previous research [9, 10] approves this claim, by showing that a relationship indeed exists between sensorimotor circuits and linguistic comprehension and production. In another study which introduces a novel VR solution for neurorehabilitation of stroke patients [19], results also show that such a system was accepted by most participants, with 44.4% strongly agreeing that the tasks offered were enjoyed, and 44.4% agreeing also.

Interestingly, both of the reviewed solutions employ a form of Rehabilitation Gaming System (RGS), which may be a possible cause for these motivating results. As defined in [20], a RGS *"is designed to engage the patients in task specific training scenarios that adapt to their performance, allowing for an individualized training of graded difficulty and complexity"*. One core aspect about RGSs is that these the models prioritise individualised training by adjusting certain parameters such as difficulty, to ensure that the patient is receiving the best treatment for their situation. Elor et al. [5], confirm the importance of having these system adjustments, by suggesting that adherence should be maintained by including parameters like difficulty and speed, thereby making the experience more comfortable for the user. Applying this information within a VR game, the proposed Project Star Catcher is shown to provide noticeably larger adherence rates compared to traditional modified constraint induced therapy [5]. An interesting recommendation made by the authors, is that different stimuli within VR rehabilitation systems, should be more salient as these are shown to go unnoticed when using minor differentiations. This motivates the need that for VR-Enhance to induce a greater sense of progression and effect, features like colour, shape and size of spells must be varied as opposed to just varying the colour, for example.

4.2 Speech Recognition

Studies have been made on the integration of speech recognition with VR to determine its practicality. An earlier investigation by Dorozhkin and Vance [21], showed that speech control was effective when applied to a VR menuing system. Stedmon et al. [22] go further in-depth and identify the indirect effects that HCI has on speech in VR systems when compared to human-to-human interaction. Their findings illustrate that when talking to a machine, in spite of participants often repeating simpler commands, more time was still needed to perform a task when compared to human interaction. The authors conclude by suggesting that for maximising VR interaction, speech should be combined with other forms of input, which VR-Enhance aims to do.

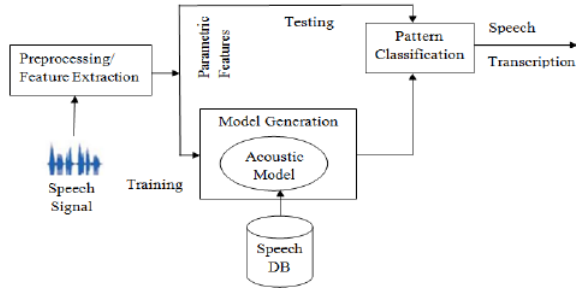


Figure 1: Speech Recognition process [23]

4.2.1 Speech Recognition Techniques

The above mentioned studies apply speech recognition in some form or another. However, these may not have been applied using the same AI methodologies. Kamble [24], provides a brief overview of the most common techniques which are applied in the STT process as seen in Figure 1. Due to differences in human speech from one person to another, feature extraction must occur to reduce such variations. As listed by Kamble, Mel Frequency Cepstrum Coefficients being robust and reliable, are preferred to Linear Predictive Coding which is another common method of feature extraction, used for lower bit rate coders. Once features are extracted and speech is pre-processed, classification algorithms are applied to recognise and transcribe the result into text. In [24], HMMs, Dynamic Time Warping and Vector Quantization are described to be the most commonly used techniques, with HMMs being the most successful. The paper however claims that the ANN has great potential in speech recognition domains.

To achieve greater results, studies also suggest hybrid models which combine techniques together. In their review [25], Cutajar et al. deduce that HMMs alone are incapable of achieving the most optimal ASR and provide examples of ANNs being integrated with HMMs. The authors also find that SVMs are provably superior to ANNs and are sometimes comparable, or even better than HMMs. The neural system-based technique proposed in [26], confirms these findings by building two models, one making use of 4 ANNs and the other based on a SVM. Results show that the SVM based model was better suited for impaired speech segmentation and both solutions outperformed segmentation techniques based on Gaussian Mixture Models and Deep Learning. CNNs have also proved their usefulness in the speech

domain. A very recent study [27] integrates this technique with a Recurrent Neural Network to provide an 8% increase in prediction accuracy from the non-hybrid CNN results. Another study [28] reports a 13.33% accuracy increase from the CNN results, through their proposed convolutional support vector machine.

4.3 Gesture Recognition

Gestures have also been applied to numerous applications for aiding medical domains. Ruppert et al [29], use the Kinect to recognise gestures and provide a contactless user interface that allowed surgeons to go through patient data during medical procedures. Phelan et al. [30], use the Oculus Rift HMD, Kinect and Myo gesture control armband to help train amputees on how to use their prosthetic arm. An earlier study also shows that children with cerebral palsy, having difficulty in muscle coordination, find VR exercises to be more fun than conventional exercises [6]. The children were also shown to complete more repetitions, hold exercises longer and have an increased range of motion when using this system. What's interesting to note, is that these solutions and those similar to them, adopt different methods of input for their gestures.

4.3.1 Gesture Input Devices

In their simulation for upper extremity rehabilitation, Gieser et al. claim that by adding more gestures when using the vision based Leap Motion, their solution could be made more feature-rich, enhancing engagement [31]. Jiang et al. argue however [32], that the number of recognisable gestures using vision based approaches is significantly less, when compared to data gloves. In brief, the authors outline how the decision of which type of gesture input device to use within a solution, ultimately depends on the application. If fast response times and accurate positioning are required, data gloves are the way to go. Otherwise, vision based approaches are a good alternative for providing a better sense of natural interaction. A gesture driven solution however, not only depends on the input device chosen, but also on the recognition technique used for gesture classification.

4.3.2 Gesture Recognition Techniques

The classification technique used, greatly depends on the approach adopted by the solution. Many recognition types exist for vision based approaches, each of which depend on a varied set of algorithmic techniques. Since VR-Enhance favours gestures captured using sensor-based controllers, the review carried is based on techniques applying this approach. Earlier forms of wearable sensor devices, take the form of accelerometers as seen in [33], which makes use of wrist bands attached to the user. The solution employs a variation of the HMM for intentional gesture recognition, which provided novel results at the time. In a later study [34], Kim et al. also make use of a HMM to provide a robust gesture recognition system for their hand-held interface. Positive results show that a more intuitive experience was provided when compared to conventional approaches.

Jung and Cha argue however, that despite being one of the most commonly used technique for gesture classification at the time, HMMs may struggle in keeping up with efficiency demands posed by gesture recognition [35]. Their ANN gesture-based solution, shows high recognition rates for five simple patterns, with suggested improvements for simultane-

ous gesture recognition. A study making use of the Wiimote controller [36], also explores the use of an ANN for generating a data set of hand gestures using their accelerometer-based recognition method. An interesting concept delivered in [37], uses a recognition algorithm based on sign sequence and template matching to achieve substantial results. The novelty in the authors' solution, lies in the fact that the method reduces the number of data points needed to represent a gesture by a significant amount. A form of this approach can also be seen in Michael's recent patent [38], which pre-processes the gesture performed into a calculated point group representing the whole gesture. The group would then be used as input to a DNN to recognise the processed gesture with exceedingly accurate results. The promising neural network technique for gesture classification is further explored with Wen et al's recent machine learning glove [14]. Using a CNN, the proposed glove is confirmed to have great potential in VR applications, achieving a recognition accuracy of 99.167% on their evaluation.

Similar to speech recognition, findings show that HMMs were initially the preferred technique for classifying gestures using sensor based approaches. These were slowly overridden by DNN architectures which have proven to be the state-of-the-art in this domain [14]. Even though gestures are capable of representing most of the individual's intent, these may sometimes limit or complicate such intentions, which may otherwise be expressed better using other modalities.

4.4 Multi-modal Interaction

As claimed by Li et al., combining gestures and speech together can intuitively express the user's intentions [18]. By having a multi-modal interface, sight impaired users would still interact with the system through hand gestures, haptic feedback and sound [39]. Alternatively, physically impaired users could make use of speech and simplified gesture patterns for also being able to use the system comfortably [39]. The benefits of using multi-modal interaction with speech and gesture modality are by far a recent discovery.

In 1980, the proposed "Put-that-there" was one of the earliest interactive systems which combined both speech and voice to enable users to generate and edit attributes of a graphic on a large display [40]. Cohen et al. established "QuickSet" in 1997, a map navigation system that supports military training through gesture and speech real time analysis. More recent studies are also shown to adopt this form of interaction, such as that seen in Chun et al.'s virtual environment [41]. A very recent paper based on fusion of speech and gesture recognition with deep learning [42], has also proven to be significantly better than a single-mode model. Emphasis was also made on an interesting approach related to classification-level fusion which combines the classification results given by the different modalities into one.

5. PROPOSED SOLUTION

Multi-modality is at core with the proposed VR-Enhance, which aims to be an immersive rehabilitation game system using VR technology with deep learning methodologies; Working simultaneously to recognise both speech and gestures, thereby expanding the range of use to not only a single therapy type, but for all the 3 types if need be.

The successes in improving adherence to VR solutions has shown to be a cause of interactive VEs. Using this knowledge, the Unity 3D engine will be used to develop a VE in

which the user will be tasked to cast spells and manipulate his surroundings by either enunciating keywords correctly, performing gestures well or a mixture of both. Maximising user independence, the patient will be shown the percentage accuracies of classification results based on how well a term was uttered or a gesture performed. This is also manifested through varying game parameters such as spell colour and amplification so that the user is encouraged to perform better to have access to more powerful spells. Indirectly, the user would be performing multiple repetitions to achieve better results similar to CT methods [2]. This information will all be represented within a diegetic interface situated within the VE for maintaining immersion and presence [43]. The solution will also be designed in a way to link spells with at least one keyword and one dynamic gesture so that the application could be customised, towards the specific needs of the patient.

Physical data gloves will be used for capturing 3D data points, as this was found to be the most accurate form of gesture input. The Valve Index VR kit will be used to compensate for the unnatural and inflexible feeling of performing gestures since this kit boasts open-hand interaction controllers that strap on to the user's hand [32]. The 3D points representing the gesture will then be pre-processed and classified by a DNN using the MiVRy service, which makes use of Michael's novel solution [38]. Using this service, the system will also cater for bimanual activities by providing recognition for both one-handed and two-handed gestures. The IBM Watson speech service will be used to convert speech into text as this provides off-the-shelf integration with Unity. Instead of applying a keyword recognising solution, the system will make use of streaming which listens to the user's speech until a pause is detected. This will attempt to mitigate the disadvantages reviewed in [22] by providing speech recognition even when the user uses a keyword within a larger sequence. Despite Watson being inferior to Google's solution in terms of recognition accuracy [44], Google's streaming service is still in beta and cannot be integrated with Unity as of yet. Additionally, when recognising dysarthric speech [45], which VR-Enhance aims to do, the proven discrepancy among the two is shown to be null; As with low speech intelligibility, the platforms are provably similar in performance.

6. EVALUATION PLAN

The efficacy of the system will be evaluated through a usability evaluation and a model evaluation of the recognition techniques. The plan will require participants to follow an on-screen script, instructing them to perform specific gestures and utter specific sequences. Expert knowledge will be gained from real-world therapists to determine the most optimal gesture motions and speech terms to use. This script would be repeated 3 times, in each case recording the relevant data for both evaluations.

6.1 Usability Evaluation

Since each spell could be cast by either doing a gesture or uttering a speech sequence, the first part will make use of classification result fusion [42], which combines the similarity values obtained after every recognition. For each user, the fused similarities will be found for every spell and from these, a global similarity representing the overall performance by the participants can be found for each playthrough. **04** will

be evaluated by comparison of the 3 globals. A post-test evaluation will also be carried out through a short survey, questioning the participants on the overall experience of the solution. Specific aspects like interface design and motivational factors will be tackled for **O3**.

6.2 Model Evaluation

Speech Recognition will be evaluated using the word error rate (WER) [46], where I words are inserted, D words deleted and S words substituted from the total amount N.

$$WER = \frac{(I + D + S)}{N} \quad (1)$$

A global WER will be calculated for every run, which is found by calculating the average of user WERs. Using the 3 derived globals, the final global mean representing the whole sitting is then found for determining the fulfillment of **O2**. Sensor-based gesture approaches are usually tested by finding the number of gestures recognised correctly, against the total number of gestures performed. Unfortunately, this does not provide enough information to properly evaluate the robustness of the model and so, Cohen's kappa (k) will be used instead [47].

$$k = \frac{p_o - p_e}{1 - p_e} \quad (2)$$

By considering the hypothetical probability of chance agreement p_e with the standard accuracy p_o the possible effects of an unbalanced data set are considered, thereby providing a more informative evaluation. A confusion matrix with gestures as its classes, will be defined for each run. Similar to speech, the mean k is found for each matrix and the global mean is then found for **O1**.

7. CONCLUSION

This report gives an overview of the proposed solution and describes how it will be evaluated. A review of studies applying VR with rehabilitation is also carried out along with different speech and gesture recognising solutions, for determining the state-of-the-art in these domains. It is believed that VR-Enhance is one of the first to leverage VR as a system for integrating real-time speech and gesture tracking, in the application of speech, occupational and physical rehabilitation.

8. REFERENCES

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